AI ASSISTANT CODING ASSIGNMENT-14.2

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Batch: 33

Semester: VI

Branch: CSE(AIML)

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Lab Objectives

• Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.

• Apply responsive, accessible, and interactive design principles.

• Create practical UI components for real-world web applications.

• Use AI to optimize layout, UX, and performance.

Task Description 1 – (Landing Page Design)

• Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

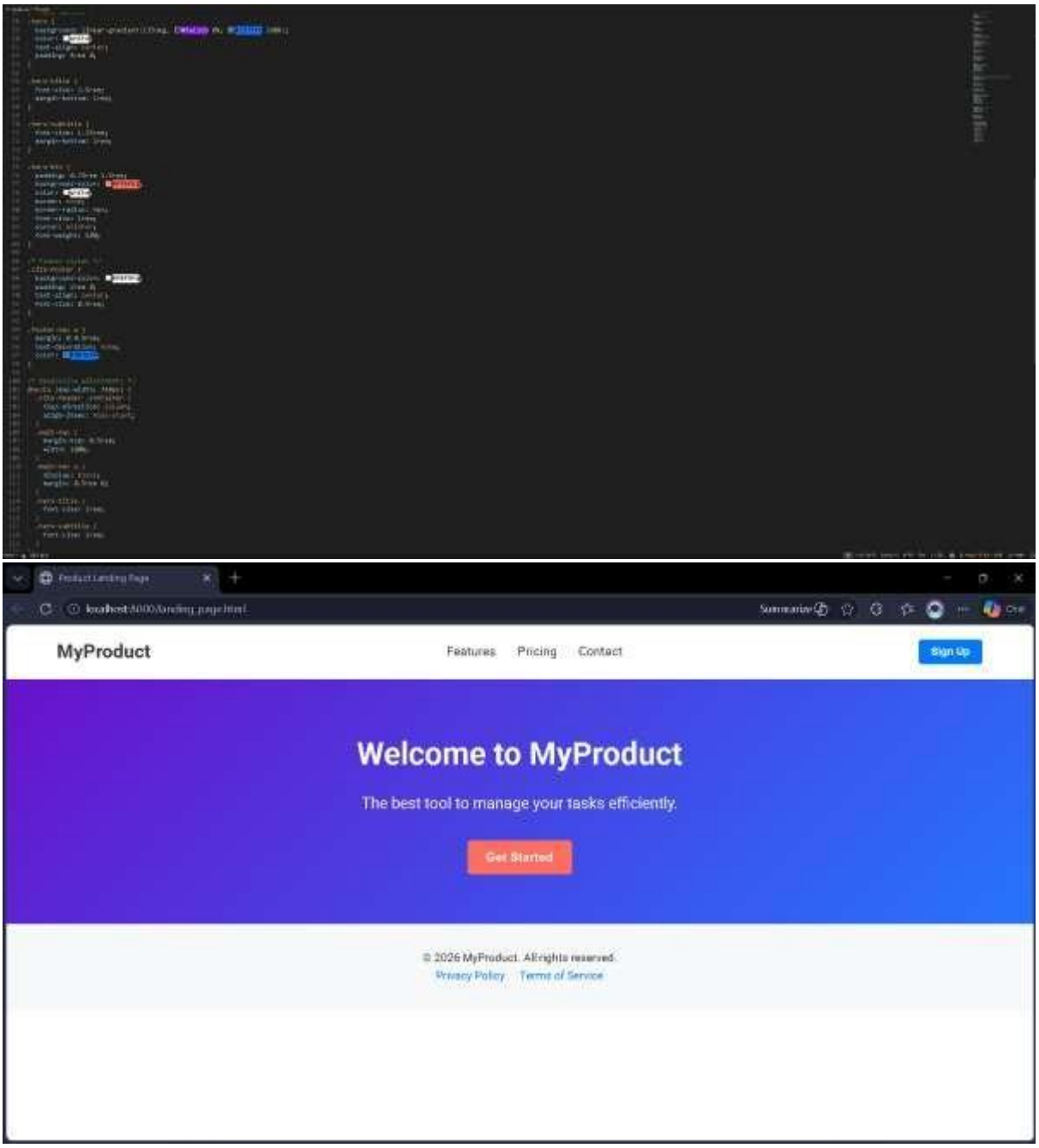
• Use HTML and CSS to structure the page.

• Use AI to suggest layout alignment and typography.

• Ensure the page is responsive across devices.

```html
<!DOCTYPE HTML>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Product Landing Page</title>
    <!-- Google Fonts for better typography -->
    <link
      href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;700&display=swap"
      rel="stylesheet"
    />

    <link rel="stylesheet" href="styles.css" />
  </head>
  <body>
    <header class="site-header">
      <div class="container">
        <div class="logo">MyProduct</div>
        <nav class="main-nav">
          <a href="#features">Features</a>
          <a href="#pricing">Pricing</a>
          <a href="#contact">Contact</a>
        </nav>
        <button class="cta-nav">Sign up free</button>
      </div>
    </header>

    <section class="hero">
      <div class="container">
        <h1 class="hero-title">Welcome to MyProduct</h1>
        <p class="hero-subtitle">
          The tool that helps you manage your tasks efficiently.
        </p>
        <button class="hero-cta">Get Started</button>
      </div>
    </section>

    <footer class="site-footer">
      <div class="container">
        <p>&copy; 2026 MyProduct. All rights reserved.</p>
        <nav class="footer-nav">
          <a href="#privacy">Privacy Policy</a>
          <a href="#terms">Terms of Service</a>
        </nav>
      </div>
    </footer>
  </body>
</html>
```

```styles.css
/* Base reset and typography */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: "Inter", sans-serif;
  line-height: 1.6;
  color: #333;
}

.container {
  width: 90%;
  max-width: 1200px;
  margin: 0 auto;
}

/* Header styles */
.site-header {
  background-color: #fff;
  border-bottom: 1px solid #eee;
}

.site-header .container {
  display: flex;
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
  padding: 1rem 0;
}

.logo {
  font-size: 1.5rem;
  font-weight: 700;
  color: #333;
}

.main-nav {
  margin: 0 0.75rem;
  text-decoration: none;
  color: #333;
  font-weight: 500;
}

.cta-nav {
  padding: 0.5rem 1rem;
  background-color: #2563eb;
  color: #fff;
  border: 0;
  border-radius: 4px;
```



Task Description 2 – (Theme Toggle Feature)

• Provide a button to switch between light and dark themes..

Requirements:

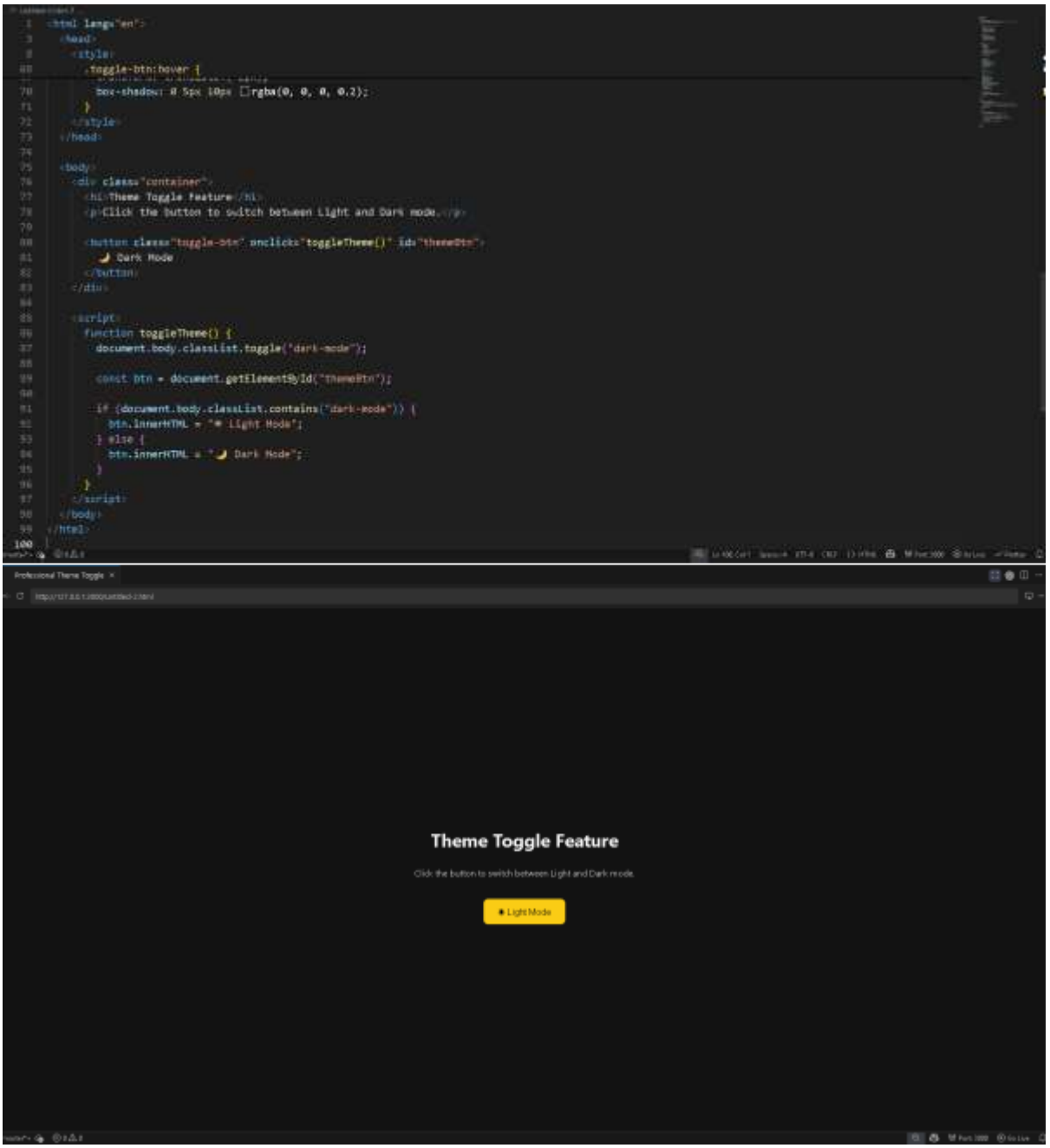
• Use JavaScript to toggle CSS classes.

• Use AI assistance to manage color contrast and accessibility.

• Ensure theme preference applies instantly on click.

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Professional Theme Toggle</title>

    <style>
      /* CSS Variables for Themes */
      :root {
        --bg-color: #ffffff;
        --text-color: #222222;
        --button-bg: #2563eb;
        --button-text: #ffffff;
      }

      body.dark-mode {
        --bg-color: #121212;
        --text-color: #f5f5f5;
        --button-bg: #facc15;
        --button-text: #000000;
      }

      /* Page Styling */
      body {
        margin: 0;
        font-family:
          Segoe UI,
          Arial,
          sans-serif;
        background: var(--bg-color);
        color: var(--text-color);
        transition:
          background 0.4s,
          color 0.4s;
```

```html
<html lang="en">
  <head>
    <style>
      /* Container */
      .container {
        display: flex;
        flex-direction: column;
        justify-content: center;
        align-items: center;
        height: 100vh;
        text-align: center;
      }

      /* Heading */
      h1 {
        margin-bottom: 10px;
      }

      /* Toggle Button */
      .toggle-btn {
        margin-top: 20px;
        padding: 12px 25px;
        font-size: 16px;
        border: none;
        border-radius: 8px;
        background: var(--button-bg);
        color: var(--button-text);
        cursor: pointer;
        transition:
          transform 0.2s,
          box-shadow 0.2s;
      }

      .toggle-btn:hover {
        transform: translateY(-2px);
```



Task Description 3 – (User Feedback Form)

• Create a feedback form with fields: Username, Rating, Comments.

Requirements:

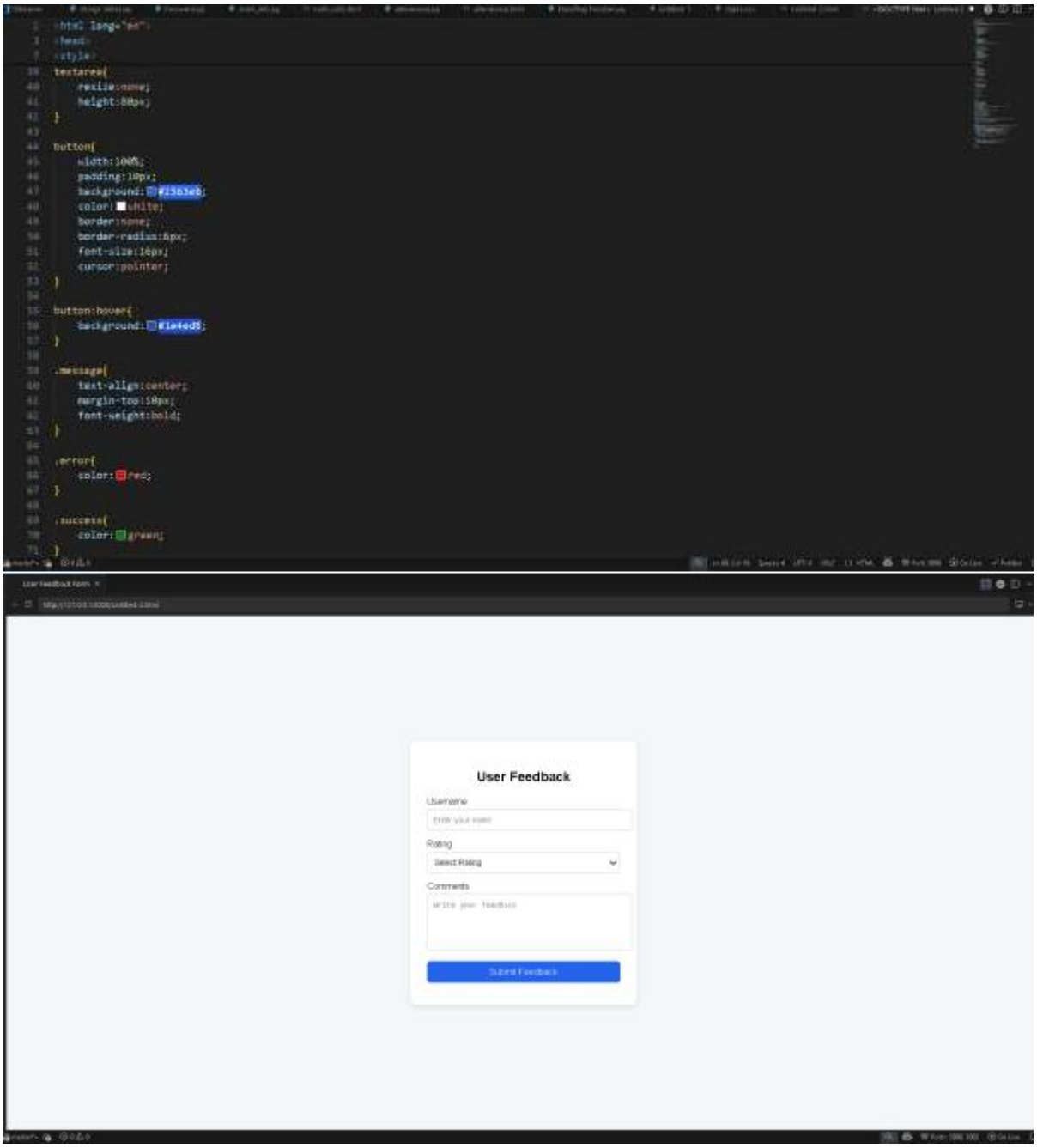
• Use JavaScript to validate required fields.

• Display success or error messages dynamically.

• Use AI to improve form usability and layout.

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>User Feedback Form</title>
<style>
body{
    font-family: Arial, sans-serif;
    background: #f4f6f8;
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
    margin:0;
}

.container{
    background: white;
    padding:30px;
    border-radius:10px;
    width:550px;
    box-shadow:0 4px 10px rgba(0,0,0,0.1);
}
h2{
    text-align:center;
    margin-bottom:20px;
}
input, select, textarea{
    width:100%;
    padding:10px;
    margin-top:5px;
    margin-bottom:15px;
    border:1px solid #ccc;
    border-radius:5px;
    font-size:14px;
```

```html
<html lang="en">
<body>
<div class="container">
<form id="FeedbackForm">
<select id="rating">
<option>4 - Very Good</option>
<option>5 - Excellent</option>
</select>
<label>Comments</label>
<textarea id="comments" placeholder="write your feedback"></textarea>
<button type="submit">Submit Feedback</button>
</form>
<div id="message" class="message"></div>
</div>
<script>
document.getElementById("feedbackForm").addEventListener("submit", function(e){
e.preventDefault();
let username = document.getElementById("username").value.trim();
let rating = document.getElementById("rating").value;
let comments = document.getElementById("comments").value.trim();
let message = document.getElementById("message");
if(username === "" || rating === "" || comments === ""){

    message.textContent = "Please fill all required fields.";
    message.className = "message error";

}
else{
    message.textContent = "Feedback submitted successfully!";
    message.className = "message success";
    document.getElementById("feedbackForm").reset();
}
});
</script>
</body>
</html>
```



Task Description - 4 (Image Gallery with JavaScript)

• Task: Display a small image gallery on a web page.

Requirements:

• Use JavaScript to dynamically load images.

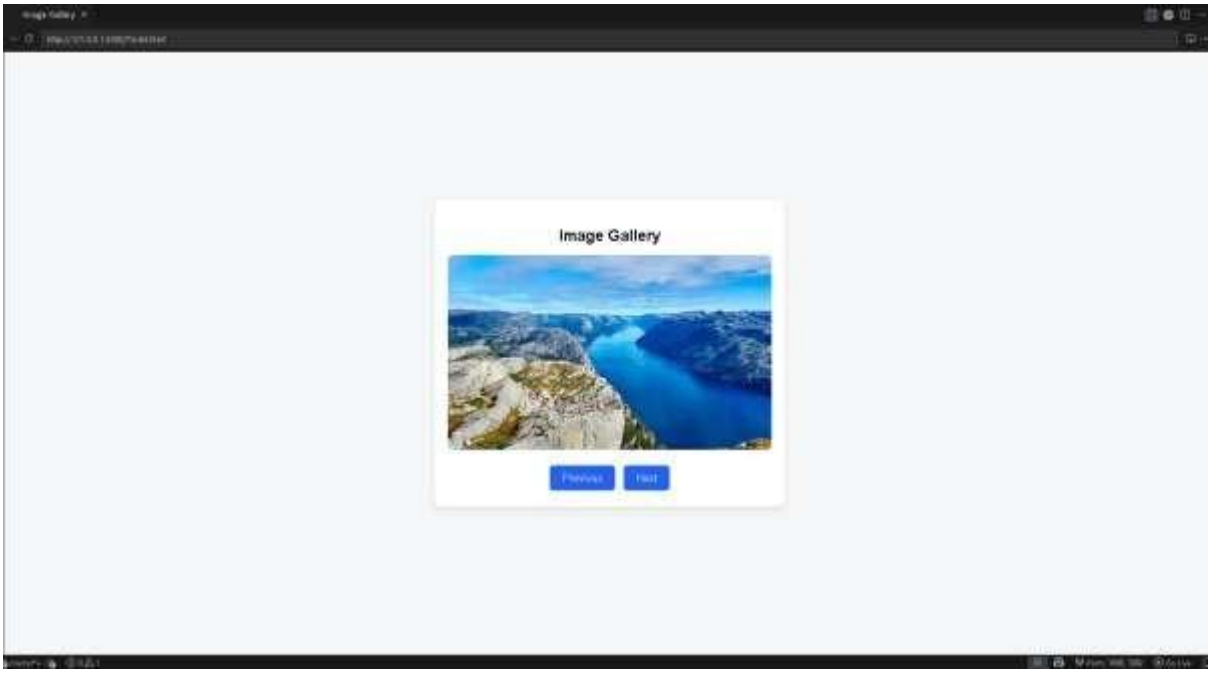
• Add next/previous navigation buttons.

• Use AI to optimize layout responsiveness.

```html
<!doctype Html>
<Html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Image Gallery</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background: #f4f6f8;
        margin: 0;
        display: flex;
        justify-content: center;
        align-items: center;
        height: 100vh;
      }
      .gallery-container {
        text-align: center;
        background: white;
        padding: 20px;
        border-radius: 10px;
        box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
        max-width: 500px;
        width: 90%;
      }
      h2 {
        margin-bottom: 15px;
      }
      img {
        width: 100%;
        height: 300px;
        object-fit: cover;
        border-radius: 8px;
      }
      .buttons {
```

```html
<Html lang="en">
  <head>
    <style>
      .buttons {
        margin-top: 15px;
      }
      button {
        padding: 10px 20px;
        margin: 5px;
        border: none;
        border-radius: 6px;
        background: #2563eb;
        color: white;
        font-size: 16px;
        cursor: pointer;
      }
      button:hover {
        background: #1e4ed8;
      }
      @media (max-width: 600px) {
        img {
          height: 200px;
        }
      }
    </style>
  </head>
  <body>
    <div class="gallery-container">
      <h2>Image Gallery</h2>
      <img id="galleryImage" src="" alt="Gallery Image" />
      <div class="buttons">
        <button onclick="prevImage()">Previous</button>
        <button onclick="nextImage()">Next</button>
      </div>
    </div>
```

```html
<html lang="en">
  <head>
  <body>
    <div class="gallery-container">
      <h2>Image Gallery</h2>
      <img id="galleryImage" src="" alt="Gallery Image" />
      <div class="buttons">
        <button onclick="prevImage()">Previous</button>
        <button onclick="nextImage()">Next</button>
      </div>
    </div>
    <script>
      const images = [
        "https://picsum.photos/id/1015/600/400",
        "https://picsum.photos/id/1025/600/400",
        "https://picsum.photos/id/1035/600/400",
        "https://picsum.photos/id/1045/600/400",
      ];
      let index = 0;
      function showImage() {
        document.getElementById("galleryImage").src = images[index];
      }
      function nextImage() {
        index = (index + 1) % images.length;
        showImage();
      }
      function prevImage() {
        index = (index - 1 + images.length) % images.length;
        showImage();
      }

      showImage();
    </script>
  </body>
</html>
```



Task Description - 5: (Notification Banner)

• Display a notification banner at the top of the page.

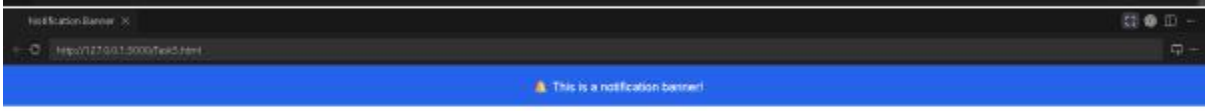
Requirements:

• Use JavaScript to show and hide the banner.

• Style the banner using CSS.

• Use AI to ensure smooth transitions and clean UI behavior.

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Notification Banner</title>
    <style>
      /* Page styling */
      body {
        margin: 0;
        font-family: Arial, sans-serif;
      }
      /* Notification banner */
      .banner {
        position: fixed;
        top: -100px;
        left: 0;
        width: 100%;
        background: #2563eb;
        color: white;
        text-align: center;
        padding: 15px;
        font-size: 16px;
        transition: top 0.5s ease;
      }
      /* when banner is visible */
      .banner.show {
        top: 0;
      }
      /* Close button */
      .close-btn {
        position: absolute;
        right: 20px;
        top: 10px;
        cursor: pointer;
```

```html
<html lang="en">
  <head>
    <style>
      .close-btn {
      }
      /* Content area */
      .container {
        padding: 120px 20px;
        text-align: center;
      }
      button {
        padding: 10px 20px;
        font-size: 16px;
        border: none;
        border-radius: 6px;
        background: #2563eb;
        color: white;
        cursor: pointer;
      }
      button:hover {
        background: #1e40af;
      }
    </style>
  </head>
  <body>
    <div id="notificationBanner" class="banner">
      🔔 This is a notification banner!
      <span class="close-btn" onclick="hideBanner()">×</span>
    </div>
    <div class="container">
      <h1>Notification Banner Example</h1>
      <p>Click the button to display the notification.</p>
      <button onclick="showBanner()">Show Notification</button>
    </div>
    <script>
```

```html
    <script>
      function showBanner() {
        document.getElementById("notificationBanner").classList.add("show");
      }
      function hideBanner() {
        document.getElementById("notificationBanner").classList.remove("show");
      }
    </script>
  </body>
</html>
```